

Geosynchronous Satellite Launch Vehicle

Geosynchronous Satellite Launch Vehicle (**GSLV**) is a class of expendable launch systems operated by ISRO. GSLV has been used in eighteen launches since 2001.

History

The Geosynchronous Satellite Launch Vehicle (GSLV) project was initiated in 1990 with the objective of acquiring an Indian launch capability for geosynchronous satellites.^{[10][11]}

GSLV uses major components that are already proven in the Polar Satellite Launch Vehicle (PSLV) launch vehicles in the form of the S125/S139 solid rocket booster and the liquid-fueled Vikas engine. Due to the thrust required for injecting the satellite in a geostationary transfer orbit (GTO) the third stage was to be powered by a LOX/LH₂ Cryogenic engine which at that time India did not possess or have the technological expertise to build. The aerodynamic characterization research was conducted at the National Aerospace Laboratories' 1.2m Trisonic Wind Tunnel Facility.^[12]

The first development flight of the GSLV (Mk I configuration) was launched on 18 April 2001 was a failure as the payload failed to reach the intended orbit parameters. The launcher was declared operational after the second development flight successfully launched the GSAT-2 satellite. During the initial years from the initial launch to 2014 the launcher had a checkered history with only 2 successful launches out of 7, resulting in the rocket gaining the nickname "naughty boy".^{[13][14]}

At the end of 2025, flight components from GSLV-F12, GSLV-F14, GSLV-F15, and GSLV-F16 are reported to still be in earth orbit as Space Junk.

Geosynchronous Satellite Launch Vehicle



GSLV-F16 during Liftoff carrying NISAR for NASA and ISRO

Function	<u>Medium-lift Launch System</u>
Manufacturer	ISRO
Country of origin	India
Cost per launch	US\$47 million ^[1]
Size	
Height	49.13 m (161.2 ft) ^[2]
Diameter	2.8 m (9 ft 2 in)

Cryogenic engine controversy

The third stage was to be procured from Russian company Glavkosmos, including transfer of technology and design details of the engine based on an agreement signed in 1991.^[11] Russia backed out of the deal after United States objected to the deal as in violation of the Missile Technology Control Regime (MTCR) in May 1992. As a result, ISRO initiated the **Cryogenic Upper Stage Project** in April 1994 and began developing its own cryogenic engine.^[15] A new agreement was signed with Russia for 7 KVD-1 cryogenic stages and 1 ground mock-up stage with no technology transfer, instead of 5 cryogenic stages along with the technology and design as per the earlier agreement.^[16] These engines were used for the initial flights and were named GSLV Mk I.^[17]

Vehicle description

The 49 m (161 ft) tall GSLV, with a lift-off mass of 415 t (408 long tons; 457 short tons), is a three-stage vehicle with solid, liquid and cryogenic stages respectively. The payload fairing, which is 7.8 m (26 ft) long and 3.4 m (11 ft) in diameter, protects the vehicle electronics and the spacecraft during its ascent through the atmosphere. It is discarded when the vehicle reaches an altitude of about 115 km (71 mi).^[18]

GSLV employs S-band telemetry and C-band transponders for enabling vehicle performance monitoring, tracking, range safety / flight safety and preliminary orbit determination. The Redundant Strap Down Inertial Navigation System/Inertial Guidance System of GSLV housed in its equipment bay guides the vehicle from lift-off to spacecraft injection. The digital auto-pilot and closed loop guidance scheme ensure the required altitude maneuver and guide injection of the spacecraft to the specified orbit.

The GSLV can place approximately 5,000 kg (11,000 lb) into an easterly low Earth orbit (LEO) or 2,500 kg (5,500 lb) (for the Mk II version) into an 18° geostationary transfer orbit.

Mass	414,750 kg (914,370 lb)
Stages	3
Capacity	
Payload to <u>LEO</u>	
Mass	6,000 kg (13,000 lb) ^[3]
Payload to <u>SSO</u>	
Mass	3,000 kg (6,600 lb) ^[3]
Payload to <u>GTO</u>	
Mass	2,500 kg (5,500 lb) ^{[2][4]}
Payload to <u>Venus</u>	
Mass	247 kg (545 lb) to 1,346 kg (2,967 lb) (direct transfer) ^[5]
Payload to <u>Mars</u>	
Mass	350 kg (770 lb) to 1,390 kg (3,060 lb) (direct transfer) ^[6]
Launch history	
Status	Mk I: Retired Mk II: Active
Launch sites	<u>Satish Dhawan Space Centre</u>
Total launches	18 + 6 (planned)
Success(es)	12
Failure	4
Partial failure	2
First flight	Mk.I: 18 April 2001 Mk.II: 15 April 2010
Last flight	Mk.I: 25 December 2010 Mk.II: 30 July 2025
Carries passengers or cargo	▪ <u>INSAT</u> ▪ <u>GSAT</u> (South Asia Satellite) ▪ <u>GISAT</u> ▪ <u>NVS</u> ▪ <u>NISAR</u>
Boosters	

Liquid boosters

The first GSLV flight, GSLV-D1 used the L40 stage. Subsequent flights of the GSLV used high pressure engines in the strap-on boosters called the L40H.^[19] The GSLV uses four L40H liquid strap-on boosters derived from the L37.5 second stage, which are loaded with 42.6 tons of hypergolic propellants (UDMH and N₂O₄). The propellants are stored in tandem in two independent tanks 2.1 m (6 ft 11 in) diameter. The engine is pump-fed and generates 760 kN (170,000 lb_f) of thrust, with a burn time of 150 seconds.

First stage

GSLV-D1 used the S125 stage which contained 125 t (123 long tons; 138 short tons) of solid propellant and had a burn time of 100 seconds. All subsequent launches have used enhanced propellant loaded S139 stage.^[19] The S139 stage is 2.8 m in diameter and has a nominal burn time of 100 seconds.^{[20][21]}

Second stage

The GS2 stage is powered by the Vikas engine. It has a diameter of 2.8 m (9 ft 2 in).^[20]

Third stage

The third stage of the GSLV Mark II is propelled by the Indian CE-7.5 cryogenic rocket engine while the older defunct Mark I is propelled using a Russian made KVD-1. It uses liquid hydrogen (LH₂) and liquid oxygen (LOX)^[22] The Indian cryogenic engine was built at the Liquid Propulsion Systems Centre.^{[23][24]} The engine has a default thrust of 75 kN (17,000 lb_f) but is capable of a maximum thrust of 93.1 kN (20,900 lb_f). In GSLV-F14 mission, a new white coloured C15 stage was introduced which has more environmental-friendly manufacturing processes, better insulation properties and the use of lightweight materials.^[25]

Pre-flight Integration of GSLV rockets

No. boosters	4 L40 Hs
Height	19.7 m (65 ft) ^[7]
Diameter	2.1 m (6 ft 11 in)
Propellant mass	42,700 kg (94,100 lb) each
Powered by	1 L40H Vikas 2
Maximum thrust	760 kN (170,000 lb _f) ^[8]
Total thrust	3,040 kN (680,000 lb _f)
Specific impulse	262 s (2.57 km/s)
Burn time	154 seconds
Propellant	N ₂ O ₄ / UDMH
First^{[8][7]} stage	
Height	20.2 m (66 ft)
Diameter	2.8 m (9 ft 2 in)
Propellant mass	138,200 kg (304,700 lb)
Powered by	1 S139 Booster
Maximum thrust	4,846.9 kN (1,089,600 lb _f)
Specific impulse	237 s (2.32 km/s)
Burn time	100 seconds
Propellant	HTPB (solid)
Second^{[8][7]} stage	
Height	11.6 m (38 ft)
Diameter	2.8 m (9 ft 2 in)
Propellant mass	39,500 kg (87,100 lb)
Powered by	1 GS2 Vikas 4
Maximum thrust	846.8 kN (190,400 lb _f)
Specific impulse	295 s (2.89 km/s)
Burn time	139 seconds
Propellant	N ₂ O ₄ / UDMH
Second GS2 (GL40)^{[8][9]} stage	
Height	11.9 m (39 ft)
Diameter	2.8 m (9 ft 2 in)
Propellant mass	42,500 kg (93,700 lb)
Powered by	1 GS2 Vikas 4
Maximum thrust	846.8 kN (190,400 lb _f)
Specific impulse	295 s (2.89 km/s)
Burn time	149 seconds



L40 booster under construction



L40 boosters and S139 core stage of GSLV F-05 during integration



GS2 stage of GSLV-F16 being integrated



GS-2 integration



GSLV F-14 Cryogenic stage being transported prior to vehicle integration



Payload fairing mating to the rocket

Propellant N_2O_4 / UDMH

Third^[7] stage (GSLV Mk I) – CUS

Height N/A

Diameter 2.8 m (9 ft 2 in)

Propellant mass N/A

Powered by 1 KVD-1

Maximum thrust 70 kN (16,000 lb_f)

Specific impulse 462 s (4.53 km/s)

Burn time N/A

Propellant LOX / LH₂

Third^[7] stage (GSLV Mk II) – CUS12

Height 8.7 m (29 ft)

Diameter 2.8 m (9 ft 2 in)

Propellant mass 12,800 kg (28,200 lb)

Powered by 1 CE-7.5

Maximum thrust 75 kN (17,000 lb_f)

Specific impulse 454 s (4.45 km/s)

Burn time 718 seconds

Propellant LOX / LH₂

Third^[9] stage (GSLV Mk II) – CUS15

Height 9.9 m (32 ft)

Diameter 2.8 m (9 ft 2 in)

Propellant mass 15,000 kg (33,000 lb)

Powered by 1 CE-7.5

Maximum thrust 75 kN (17,000 lb_f)

Specific impulse 454 s (4.45 km/s)

Burn time 846 seconds

Propellant LOX / LH₂

Variants

GSLV rockets using the Russian Cryogenic Stage (CS) are designated as the GSLV Mark I while versions using the indigenous Cryogenic Upper Stage (CUS) are designated the GSLV Mark II.^{[26][27]} All GSLV launches have been conducted from the Satish Dhawan Space Centre in Sriharikota.

GSLV Mark I

The first developmental flight of GSLV Mark I had a 129 tonne (S125) first stage and was capable of launching around 1500 kg into geostationary transfer orbit. The second developmental flight replaced the S125 stage with S139. It used the same solid motor with 138 tonne propellant loading. The chamber pressure in all liquid engines were enhanced, enabling a higher propellant mass and burn time. These improvements allowed GSLV to carry an additional 300 kg of payload.^{[28][29]} The fourth operational flight of GSLV Mark I, GSLV-F06, had a longer third stage called the C15 with 15 tonne propellant loading and also employed a 4 meter diameter payload fairing.^{[30][31]}

GSLV Mark II

This variant uses an Indian cryogenic engine, the CE-7.5, and is capable of launching 2500 kg into geostationary transfer orbit. Previous GSLV vehicles (GSLV Mark I) have used Russian cryogenic engines.^[32]

For launches from 2018, a 6% increased thrust version of the Vikas engine was developed. It was demonstrated on 29 March 2018 in the GSAT-6A launch second stage. It will be used for the four Vikas engines on the first stage boosters on future missions.^[33]

A 4m diameter Ogive payload fairing was developed and deployed for the first time in the EOS-03 launch on 12 August 2021, although this launch was a failure due to technical anomalies with the Cryogenic Upper Stage. This will allow GSLV vehicles to accommodate larger payloads.^[34]

As of October 2024, ISRO has stopped selling GSLV Mk II Rockets. Eight known launches are planned with NVS Missions, IDRSS Missions, NISAR Mission.^{[35][36]}

RLV-OREX

The Reusable Launch Vehicle Technology Demonstration program, is a prototype spaceplane concept created by ISRO. For the Orbital return Flight experiment, a modified version of the GSLV Mk. II launcher, with the upper Cryogenic Stage replaced with the PS-4 stage from the PSLV is currently in development, as the RLV won't need all the excess energy produced by the CUS.^{[37][38]}

Launch statistics

Launch system status

- ☒ Active
- ☐ Retired

Variant	Launches	Successes	Failures	Partial failures
GSLV Mk. I	6	2	2	2
GSLV Mk. II	12	10	2	0
Total as of July 2025 ^[39]	18	12	4	2

Decade-wise summary of GSLV Launches

Decade	Successful	Partial success	Failure	Total
2000s	2	2	1	5
2010s	6	0	2	8
2020s	4	0	1	5
Total	12	2	4	18

Gallery



GSLV F05 take off



GSLV F11 vehicle at Second Launch Pad



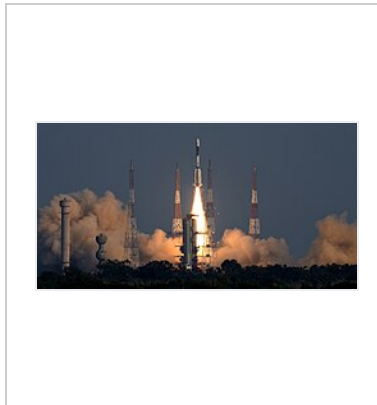
Indigenous Cryogenic Upper Stage CE-7.5 of GSLV



Fully integrated GSLV-F05 coming out of the Vehicle Assembly Building

Launch of GSLV F11 from Second Launch Pad

Top view of a fully Integrated GSLV-F08 inside the Vehicle Assembly Building



GSLV F11 launch

See also

- [SSLV](#)
- [Comparison of orbital launchers families](#)
- [Comparison of orbital launch systems](#)
- [PSLV](#)
- [LVM3](#)
- [NGLV](#)
- [ISRO](#)
- [ASLV](#)
- [SLV](#)



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